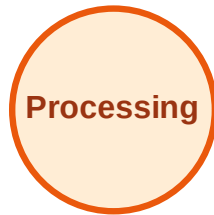


BFS Traversal

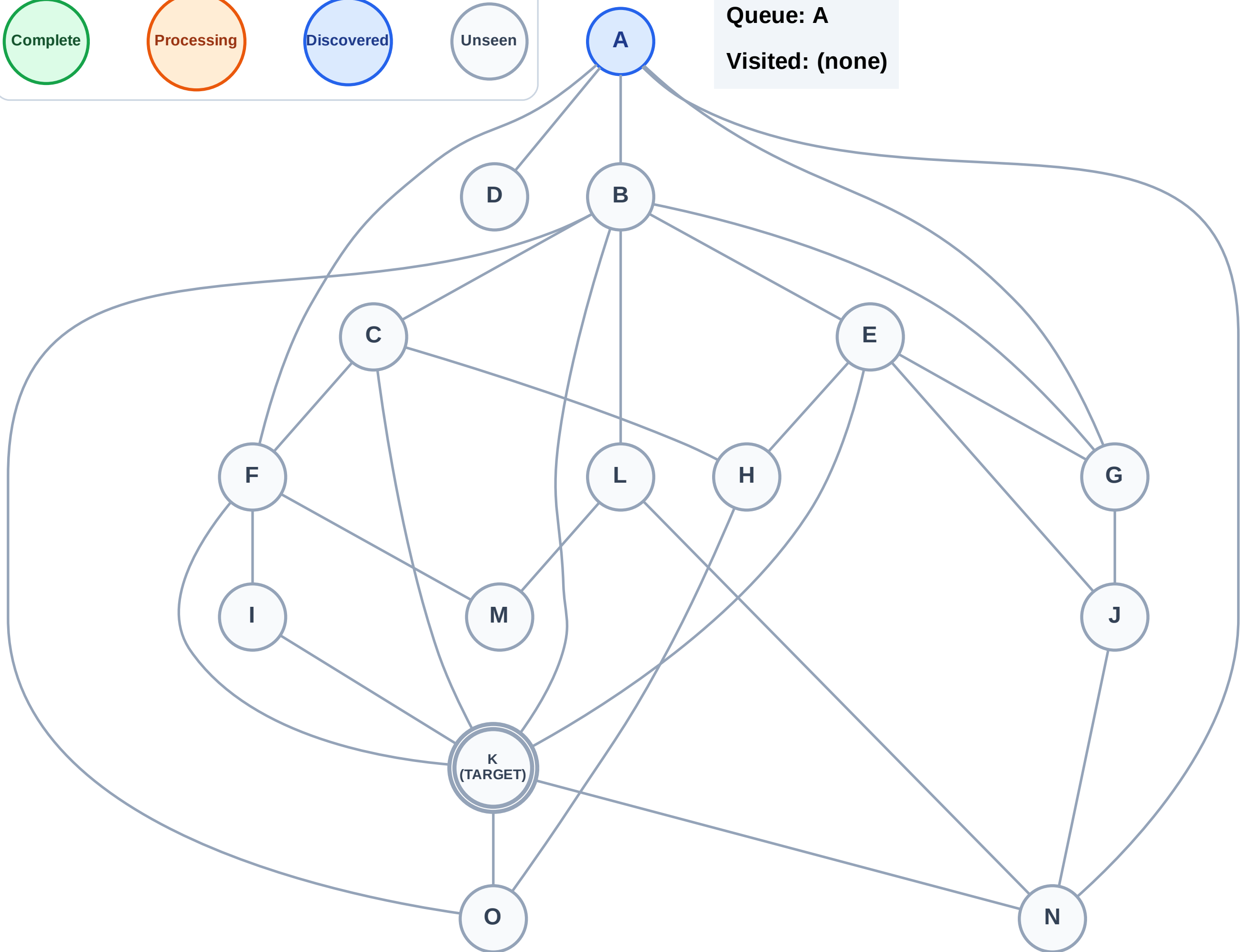
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

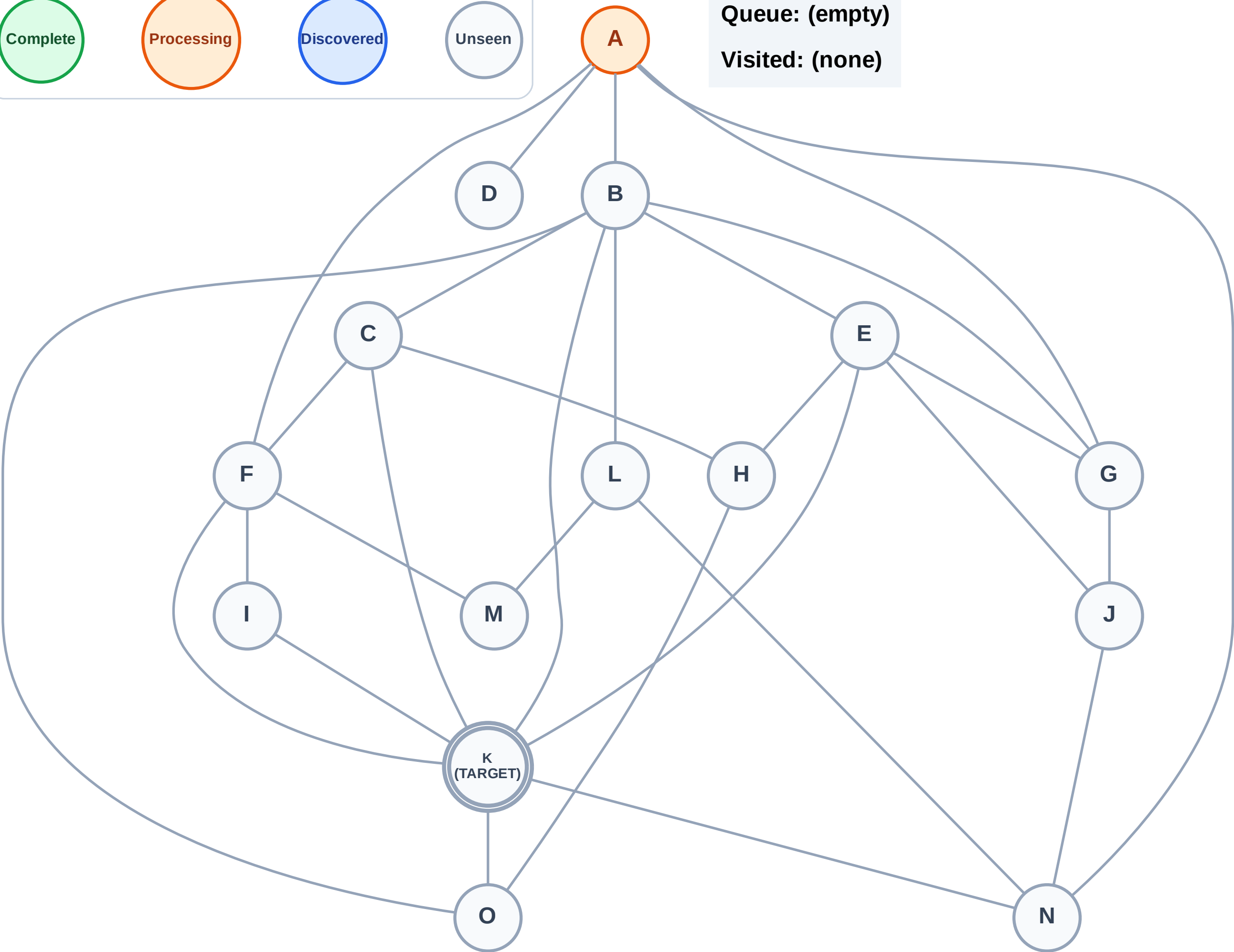
Complete

Processing

Discovered

Unseen

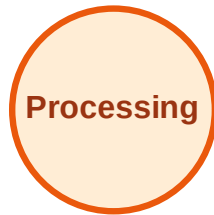
Queue: (empty)
Visited: (none)



BFS Traversal

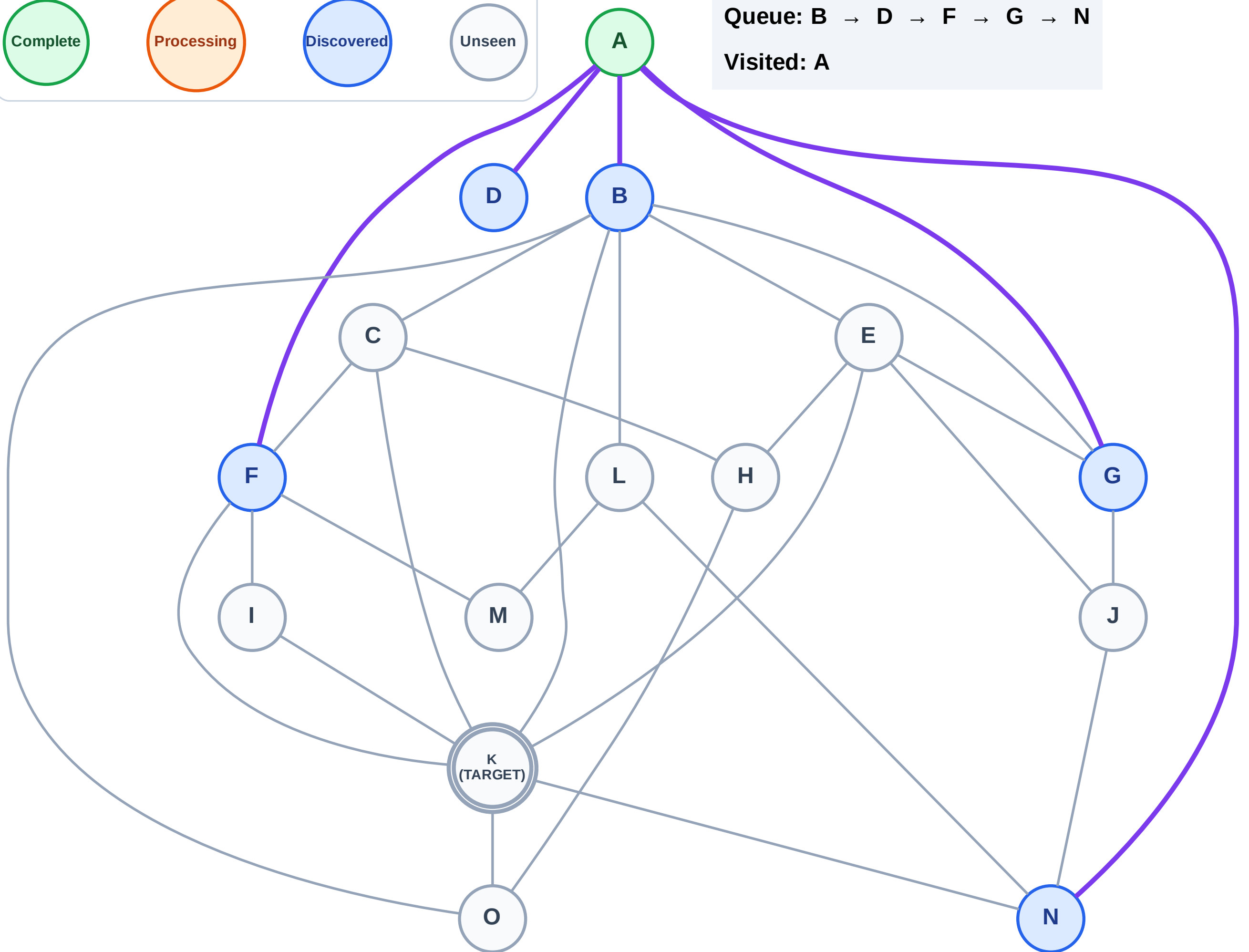
Step 3: Discover B, D, F, G, N from A

Legend



Queue: B → D → F → G → N

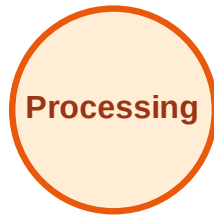
Visited: A



BFS Traversal

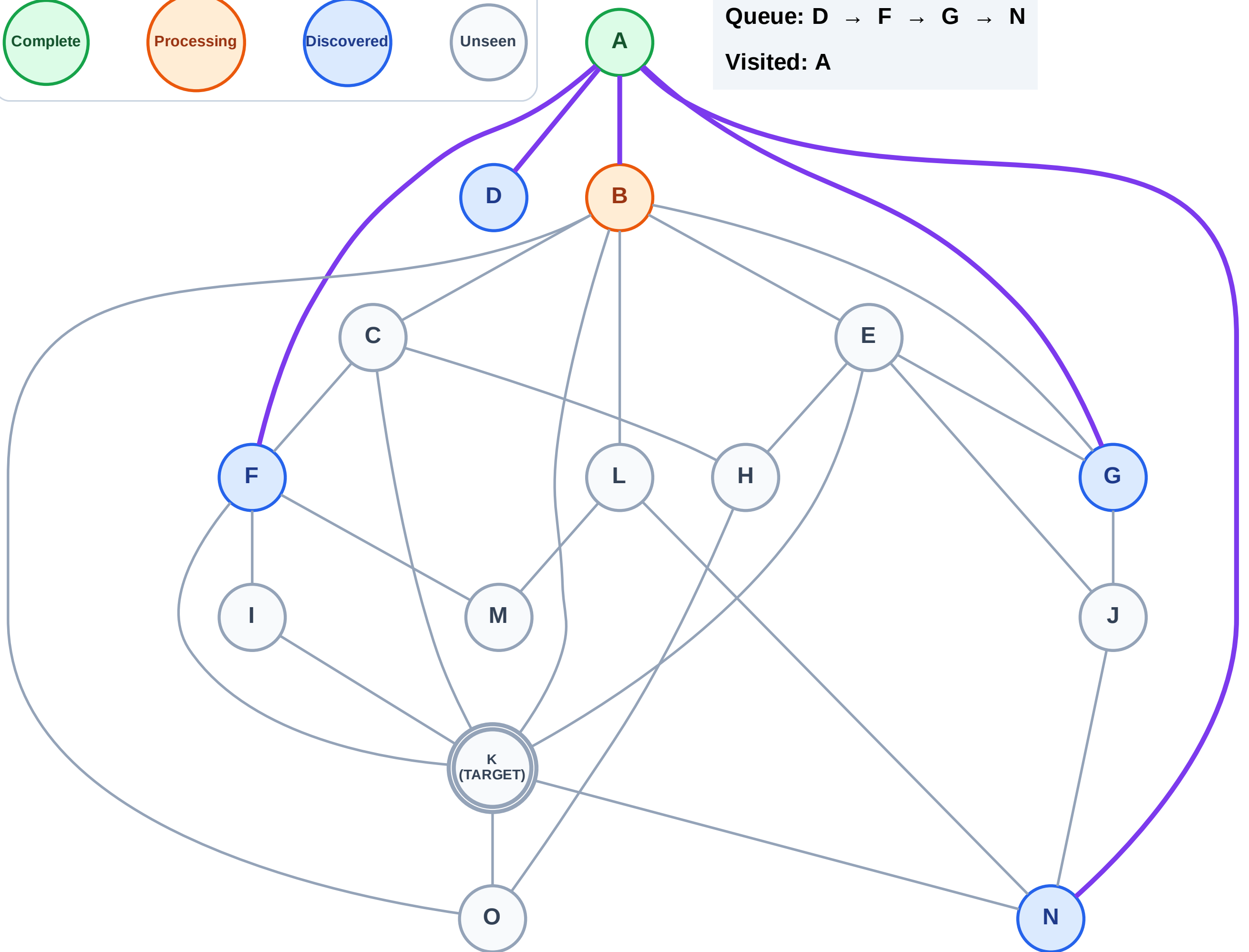
Step 4: Process node B

Legend



Queue: D → F → G → N

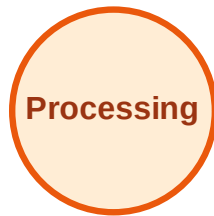
Visited: A



BFS Traversal

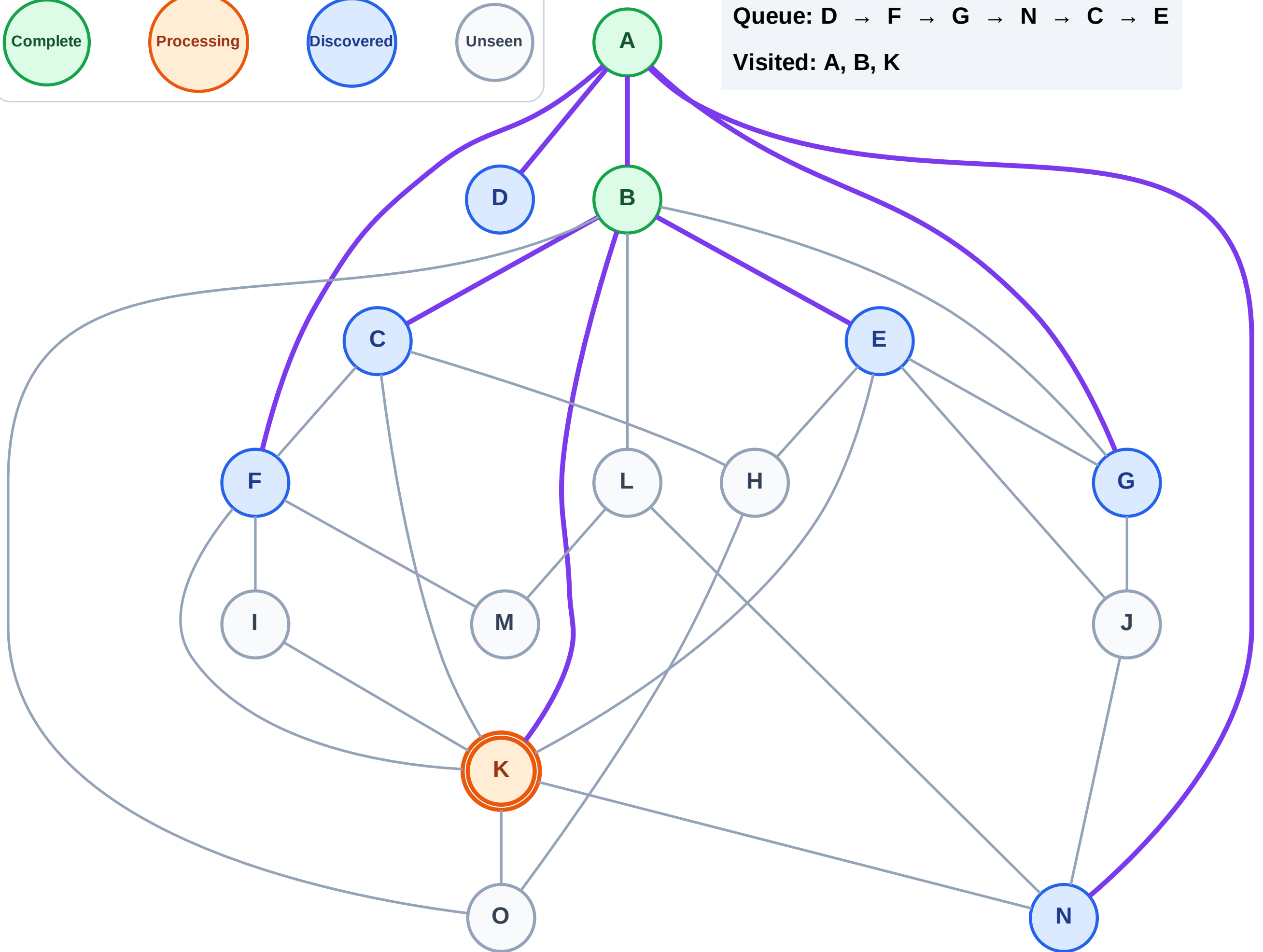
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: D → F → G → N → C → E

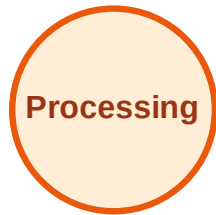
Visited: A, B, K



DFS Traversal

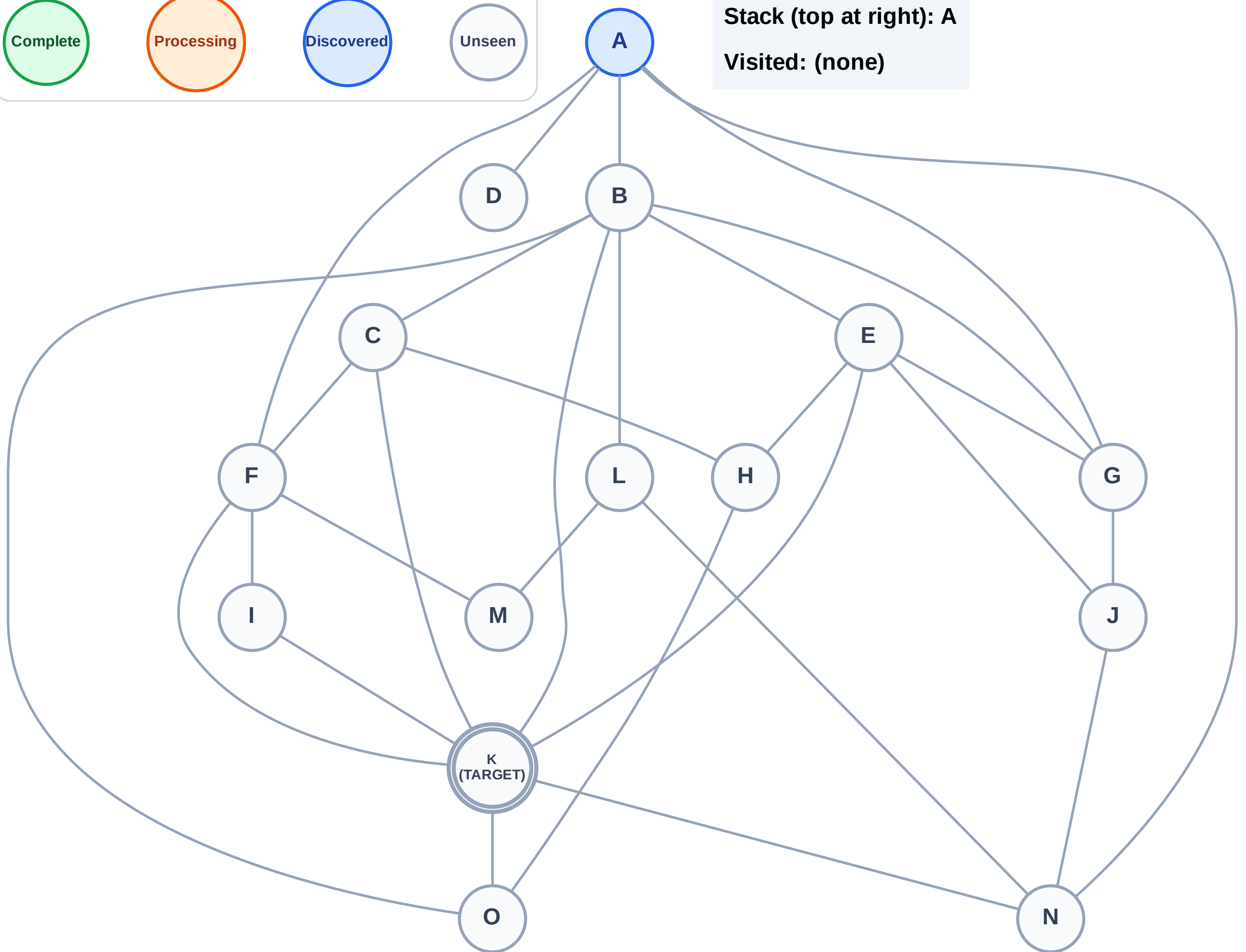
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

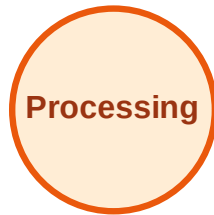
Visited: (none)



DFS Traversal

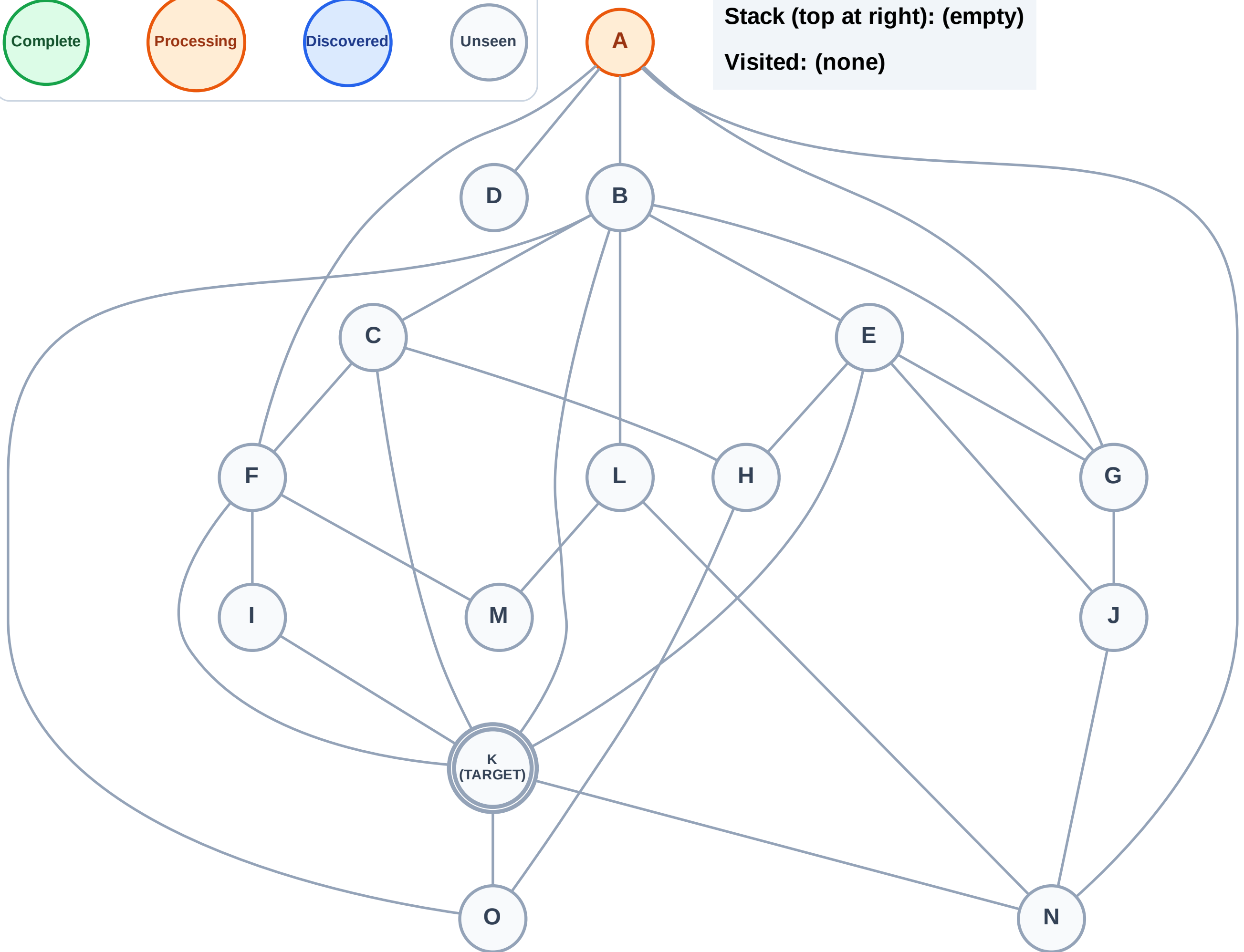
Step 2: Process node A

Legend



Stack (top at right): (empty)

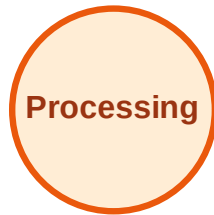
Visited: (none)



DFS Traversal

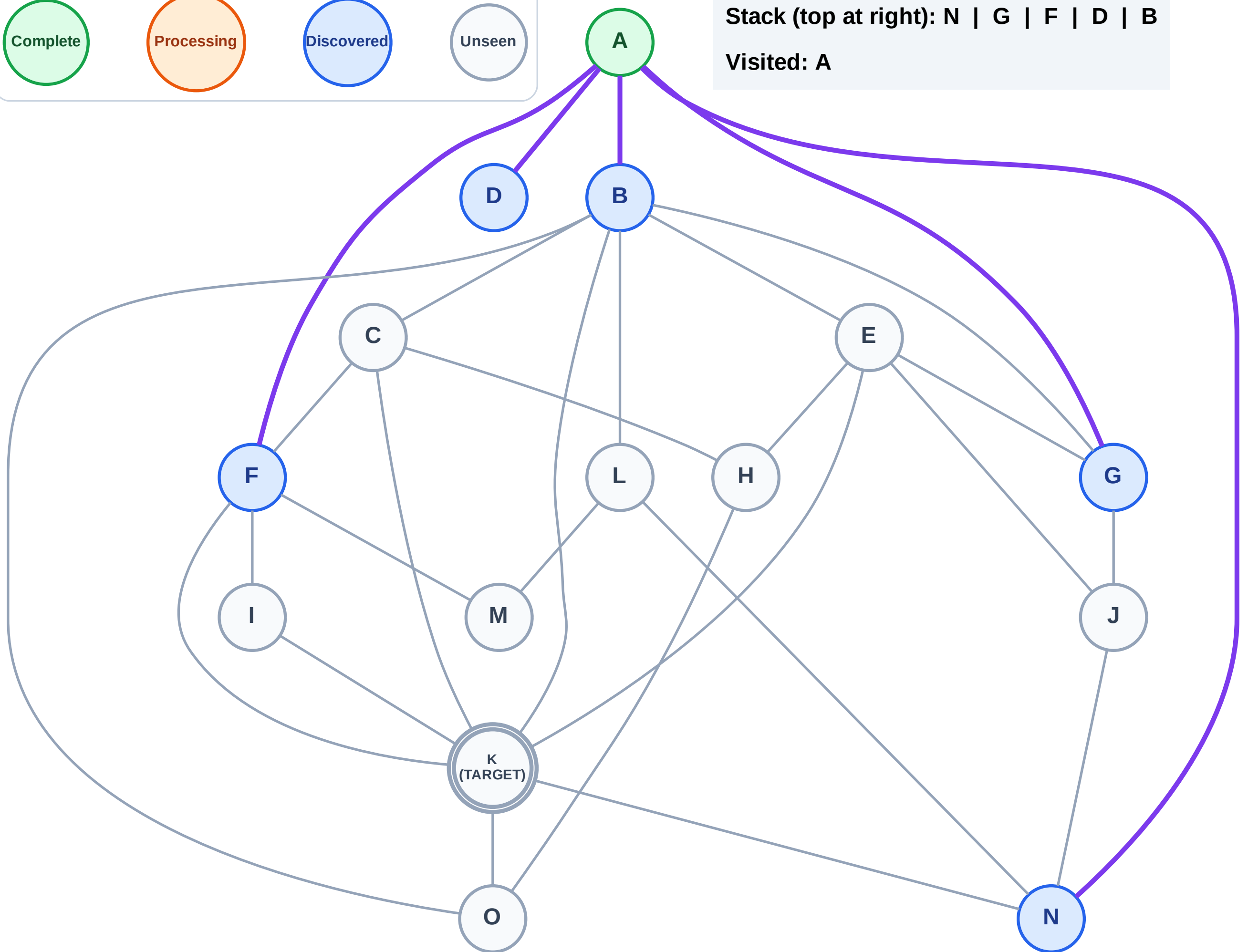
Step 3: Discover N, G, F, D, B from A

Legend



Stack (top at right): N | G | F | D | B

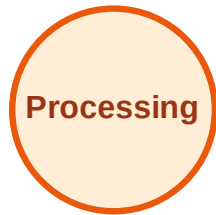
Visited: A



DFS Traversal

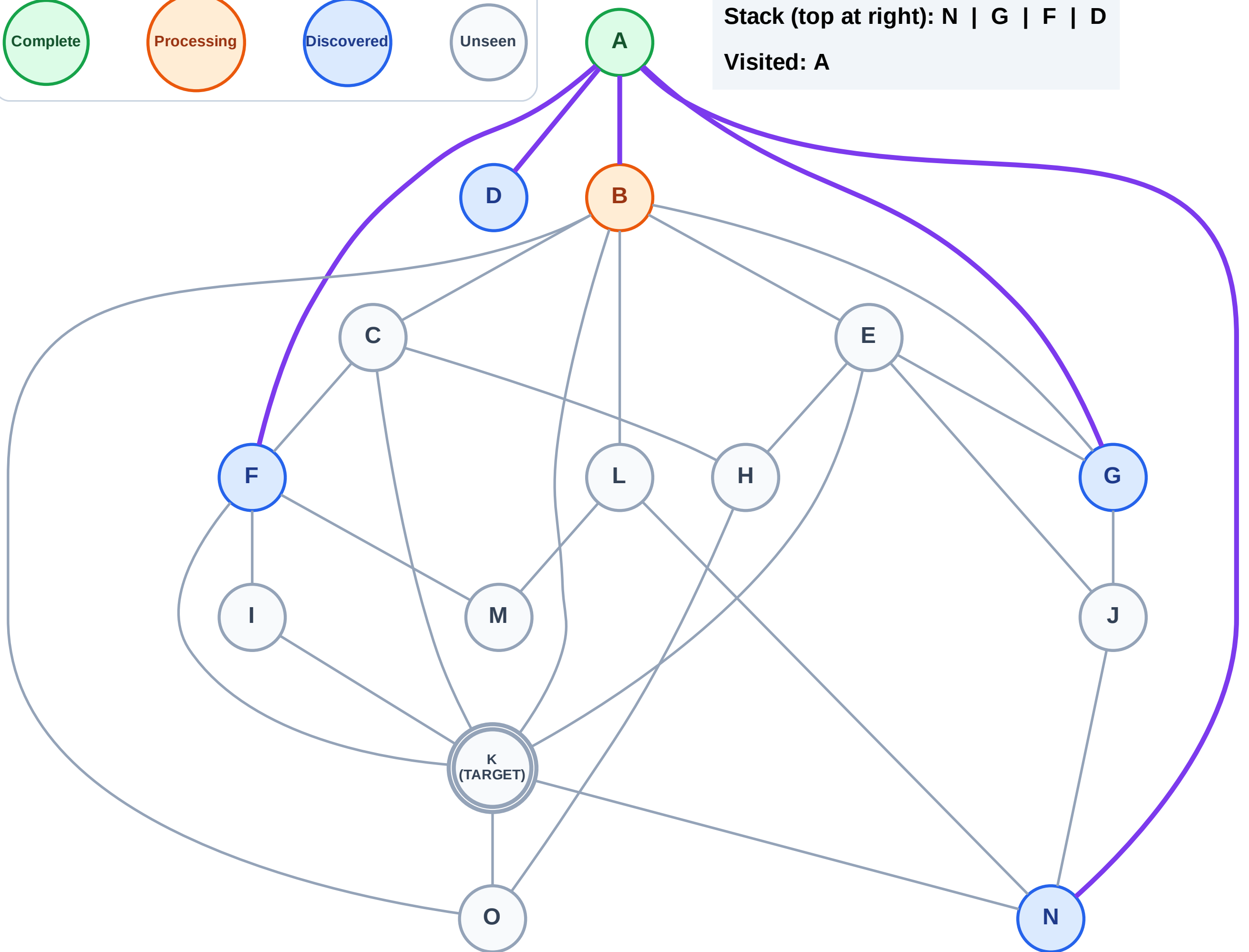
Step 4: Process node B

Legend



Stack (top at right): N | G | F | D

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): $A \rightarrow B \rightarrow K$

